genes stand to gain from the snail's reproduction, fluke genes don't. This is because any given fluke has no particular expectation that its genes will be housed in its present host's offspring. They might be, but so might those of any of its fluke rivals. Given that snail longevity has to be bought at the cost of some loss in the snail's reproductive success, the fluke genes are 'happy' to make the snail pay that cost, since they have no interest in the snail's reproducing itself. The snail genes are not happy to pay that cost, since their long-term future depends upon the snail reproducing. So, I suggest that fluke genes exert an influence on the shell-secreting cells of the snail, an influence that benefits themselves but is costly to the snail's genes. This theory is testable, though it hasn't been tested yet.

We are now in a position to generalize the lesson of the caddises. If I am right about what the fluke genes are doing, it follows that we can legitimately speak of fluke genes as influencing snail bodies, in just the same sense as snail genes influence snail bodies. It is as if the genes reached outside their 'own' body and manipulated the world outside. As in the case of the caddises, this language might make geneticists uneasy. They are accustomed to the effects of a gene being limited to the body in which it sits. But, again as in the case of the caddises, a close look at what geneticists ever mean by a gene having 'effects' shows that such uneasiness is misplaced. We need to accept only that the change in snail shell is a fluke adaptation. If it is, it has to have come about by Darwinian selection of fluke genes. We have demonstrated that the phenotypic effects of a gene can extend, not only to inanimate objects like stones, but to 'other' living bodies too.

The story of the snails and flukes is only the beginning. Parasites of all types have long been known to exert fascinatingly insidious influences on their hosts. A species of microscopic protozoan parasite called *Nosema*, which infests the larvae of flour beetles, has 'discovered' how to manufacture a chemical that is very special for the beetles. Like other insects, these beetles have a hormone called the juvenile hormone which keeps larvae as larvae. The normal change from larva to adult is triggered by the larva ceasing production of juvenile hormone. The parasite *Nosema* has succeeded in synthesizing (a close chemical analogue of) this hormone. Millions of *Nosema* club together to mass-produce juvenile hormone in the beetle larva's body, thereby preventing it from turning into an adult. Instead it goes on growing, ending up as a